ALICE L WALTON SCHOOL OF MEDICINE

DOACs

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Adapted from: Anticoagulant Drugs

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0:00 / 11:08 1x

Learning Objectives (3) **Versions**

After completing this brick, you will be able to:

1. Justify the use of pharmacological agents in treating DVT and PE based on their mechanism of action and pharmacokinetic profile.
2. Recognize adverse drug reactions for medications used for DVT and PE.
3. Apply an evidence-based pharmacotherapeutic approach to treat DVT and PE.

CASE CONNECTION

HT is a 43-year-old female who comes to the emergency department with reports of right calf swelling. "It started swelling yesterday," she reports. "And it really hurts when I touch it." When you ask if she has traveled lately, she replies, "I just got back from Australia two days ago. It was a great trip, but the flight was long and brutal." HT's vital signs are within normal limits. A physical exam is remarkable for right calf erythema, swelling, and tenderness. You order a right lower-extremity venous ultrasound, which reveals a thrombosis in the femoral vein.

How will you proceed with HT's treatment? Consider your answers as you read, and we'll revisit HT at the end of the brick.

GO TO CONCLUSION ↓

In your previous coursework (Thrombotic Disorders: Foundations and Frameworks) we have talked about three of the four anticoagulant medication classes. We discussed the clotting factor synthesis inhibitor (vitamin K antagonist) warfarin, the heparins which are antithrombin activators, and the direct thrombin inhibitors like argatroban. In this brick we will learn about the newest class of anticoagulants, the *direct factor Xa inhibitors*, or more commonly referred to as direct oral anticoagulants (DOACs).

How do DOACs work? (LO#1)

Direct factor Xa inhibitors block factor Xa's action in the common pathway and therefore inhibit coagulation (Figure 1). Available drugs include apixaban, rivaroxaban, betrixaban, and edoxaban and are also called **direct oral anticoagulants** (DOACs). They are only administered *orally* and directly inhibit a component of the coagulation pathway, unlike the indirect actions of heparin and warfarin.

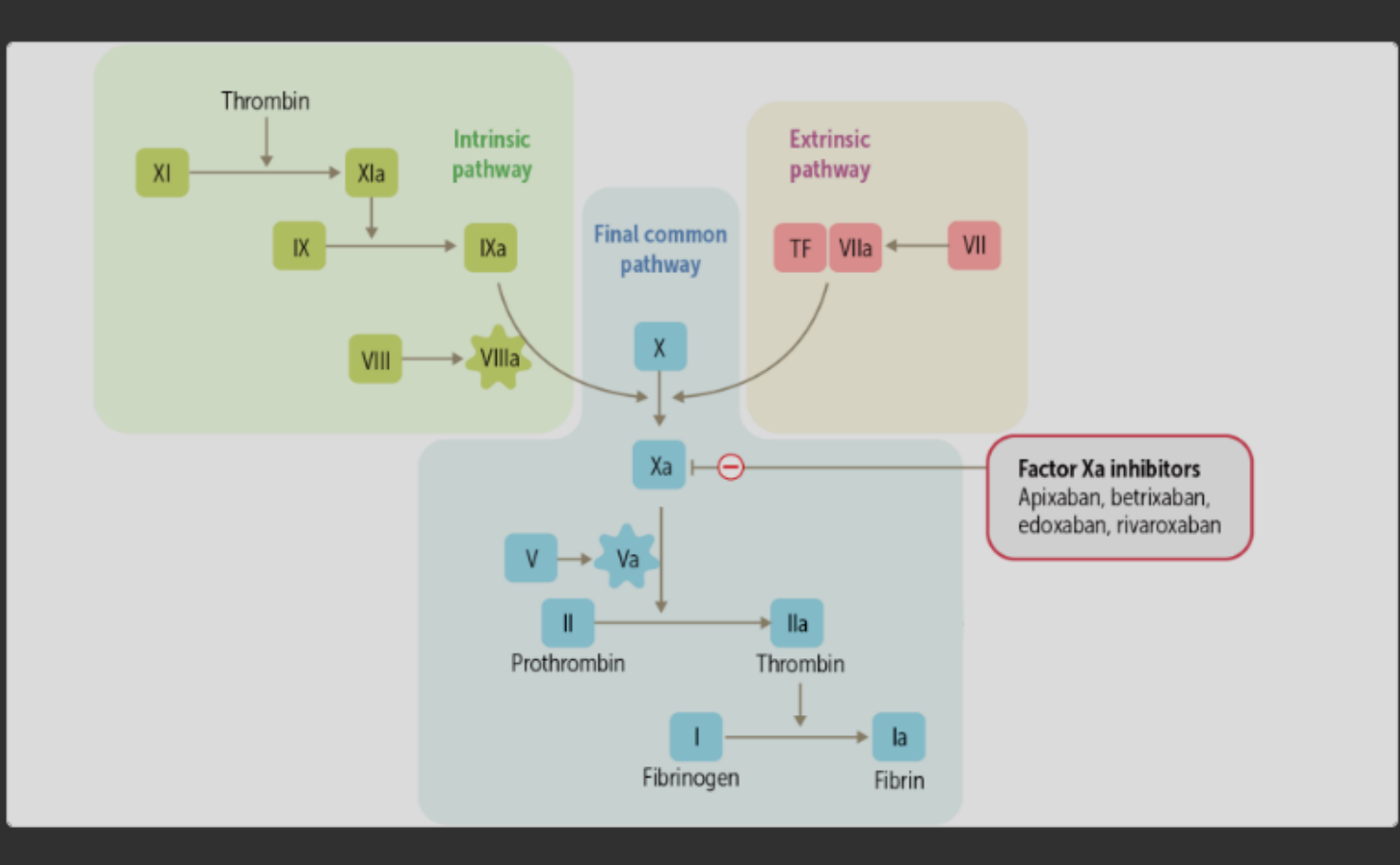


Figure 1

Indications (LO#3)

Since these medications are taken orally, DOACs can be used for almost all the same indications as for warfarin, including treatment of VTE and preventing emboli in patients wi[illegible]ed for preventing DVTs after a major surgery. *Where they differ from warfarin, in terms of indication, is that they are NOT indicated for anticoagulation in patients with a prosthetic heart valve.*

Mnemonic

Remember the direct Xa inhibitors: **Ban** on factor **Xa**: api**XaBAN**, rivaro**XaBAN,** betri**XaBAN**, edo**XaBAN**

Drug Monitoring (LO#3)

A benefit of DOACs, over other anticoagulants like warfarin, is that they do not require frequent monitoring of PT/INR. Just general patient monitoring for signs and symptoms of irregular bleeding (e.g., overt bruising, dark/ tarry stools, etc.) is needed with these agents.

Adverse Effects (LO#2)

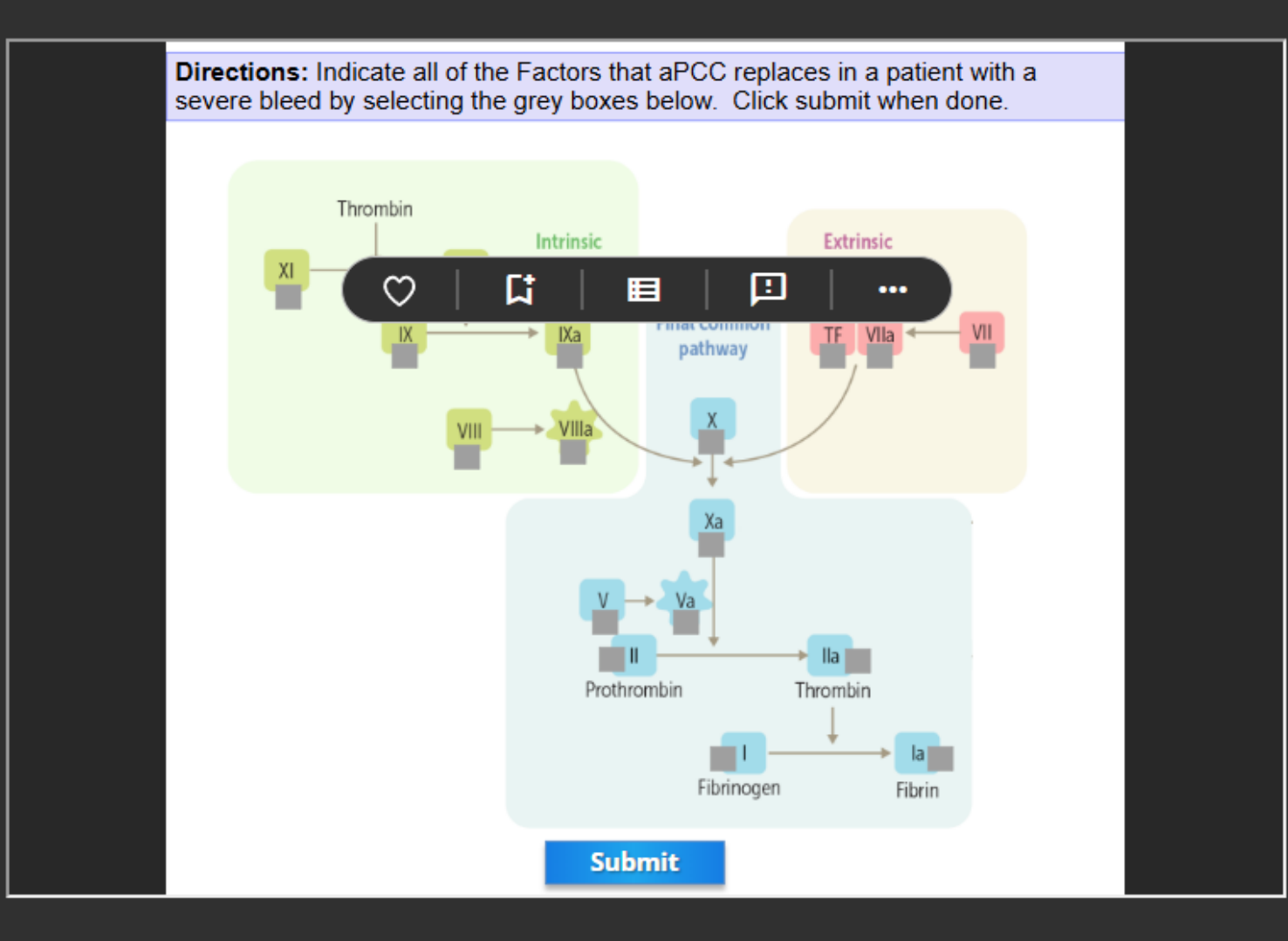
Just like the other ant[illegible]ssociated with DOACs is unwanted/ severe bleeding. If this does occur there are a couple of strategies to reverse DOAC induced bleeding.

First, and most commonly, activated prothrombin complex concentrate (aPCC) is used as soon as the severe/ life threatening bleeding is recognized. As we have previously discussed with the other anticoagulant medications, aPCC replaces clotting factors (II, VII, IX, and X) to aid the patient in supplementing the clotting cascade (Figure 2). Giving the patient aPCC and replacing clotting factors does NOT inactivate the DOAC medication. The second strategy is a targeted reversal agent.

Andexanet alfa is a specific reversal agent for the DOACs rivaroxaban and apixaban, by far the two most common medications in the class. It binds and sequesters these DOACs and inhibits the activity of Tissue Factor Pathway Inhibitor, increasing tissue factor-initiated thrombin generation.

CLINICAL CORRELATION

Andexanet alfa seems ideal. To have a specific reversal agent for DOACs could be very beneficial, however, significantly higher *drug cost* and *failure to show superiority* in clinical trials have limited the use of andexanet alfa.



Directions: Indicate all of the Factors that aPCC replaces in a patient with a severe bleed by selecting the grey boxes below. Click submit when done.

Figure 2

Which DOACs are reversed by andexanet alfa?

What are the contraindications and drug interactions? (LO#2)

DOACs are contraindicated in patients with active, heavy bleeding or for those with severe allergies to the drugs. Safety during pregnancy is not established, so most sources recommend not using the drugs during pregnancy.

All of the DOACs are eliminated renally, so dosage adjustment for patients with altered renal function is required.

The DOACs are also substrates of CYP3A4, so caution should be used when other medications that induce/ inhibit this enzyme are used. Particularly, the inhibitors because it greatly increases the risk for bleeding. The DOACs should also be used cautiously in patients with severe liver disease because of the hepatic metabolism.

Which part of the coagulation pathway is inhibited by direct factor Xa inhibitors?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to HT, how do you proceed with treatment?

You explain to HT that a clot has formed in one of the veins of her right leg, likely because of prolonged sitting on the airplane. The main danger of this is that a piece of the clot may break off and travel to the lungs. You recommend initiating treatment immediately with a direct factor Xa anticoagulant, like apixaban. The DOACs are a convenient oral option, and HT is a good candidate for treatment with one of these medications. HT does well during treatment and completes the full course without bleeding complications.

Summary

- Direct factor Xa inhibitors, also known as direct oral anticoagulants (DOACs), block factor Xa in the coagulation pathway, preventing clot formation.
- DOACs are administered orally and are used for conditions like VTE, atrial fibrillation, and post-surgery DVT prevention, but NOT for patients with prosthetic heart valves.
- Unlike warfarin, DOACs do not require frequent PT/INR monitoring, but general monitoring for signs of bleeding is necessary.
- The main adverse effect of DOACs is severe bleeding, which can be managed with activated prothrombin complex concentrate (aPCC) or andexanet alfa.
- DOACs are contraindicated in patients with active bleeding, severe allergies, and du[illegible]djustments for renal impairment and caution with CYP3A4 inhibitors.
- Direct Xa inhibitors like apixaban and rivaroxaban are alternatives to warfarin, directly targeting factor Xa in the common coagulation pathway.

Review Questions

1. Which clotting factor is inhibited by the DOAC medication class?

A. ○ IIa
B. ○ VII
C. ○ IXa
D. ● Xa

Hide Explanation

The correct answer is factor Xa (D). DOACs directly inhibit clotting factor Xa in the common pathway to inhibit coagulation. Factor IIa (A), or thrombin, is inhibited by direct thrombin inhibitors, but not DOACs. Factor VII (B) and IXa (C) are from the extrinsic and intrinsic pathways, respectively, and not inhibited by DOACs.

2. A patient is given rivaroxaban to treat deep vein thrombosis and subsequently develops heavy gastrointestinal bleeding with hypotension. Which of the follo[illegible] heparin in this patient?

A. ○ Antithrombin
B. ● Activated prothrombin complex concentrate
C. ○ Protamine sulfate
D. ○ Tranexamic acid
E. ○ Vitamin K

Hide Explanation

The correct answer is Activated prothrombin complex concentrate (aPCC) (B). It contains clotting factors II, VII, IX, and X that help reverse the effects of rivaroxaban. Antithrombin (A) is an endogenous anticoagulant that is activated by heparin. Protamine sulfate (C) is an antidote to bleeds caused by heparin. Tranexamic acid (D) is an antidote for thrombolytic drugs. Vitamin K (E) is also an antidote to bleeds caused by warfarin.

3. The DOACs are substrates of which enzyme?

A. ○ CYP2D6
B. ● CYP3A4
C. ○ CYP2C9
D. ○ P-glycoprotein

Hide Explanation

The correct answer is CYP3A4 (B). DOACs are substrates of CYP3A4, so caution is needed with medications that induce/inhibit this enzyme. CYP2D6 (A) and CYP2C9 (C) are other cytochrome P450 enzymes but not involved with DOACs. P-glycoprotein (D) is a transporter protein, not an enzyme.

How would you rate this Brick?

References

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Justify the use of pharmacological agents in treating DVT and PE based on their mechanism of action and pharmacokinetic profile.~~
02 ☑ ~~Recognize adverse drug reactions for medications used for DVT and PE.~~
03 ☑ ~~Apply an evidence-based pharmacotherapeutic approach to treat DVT and PE.~~

Objective Confidence: 3/3